

DP Computer Science: A 1.2.1 Representing Data — Number Systems

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

1. State the **base**, **digit set**, and **place-value rule** for binary and for hexadecimal, and describe how each represents a whole number
2. Convert an integer between **binary and decimal** in both directions, showing each step
3. Convert an integer between **hexadecimal and decimal** in both directions, showing each step
4. Convert an integer between **binary and hexadecimal** in both directions using **four-bit grouping**
5. Check a conversion by **reversing it**, and select the method they find most reliable

Command Term Note: This lesson is held at **Describe**—a detailed account. Fractional place value has been cut because the statement asks only for whole numbers. Text, images, sound, and video follow in A1.2.2.

Key Vocabulary (Tier 3)

Term	Definition
Bit	A single binary digit, either 0 or 1
Nibble	A group of four bits, which is exactly one hexadecimal digit
Byte	A group of eight bits
Base (radix)	The number of digit symbols a system uses: 2 for binary, 16 for hexadecimal, 10 for decimal
Place value	The value of a position, each one a fixed multiple of the position to its right
Hexadecimal	A base-16 system using 0–9 then A–F, where one digit stands for four bits

Pedagogical Focus

- **Self-Management Skills (ATL):** Students consciously practise self-monitoring: they check each conversion by reversing it, log which answers they got right first time, and decide which of the two methods they will rely on under exam conditions.
- **Assessment (ATT):** The lesson runs on visible evidence rather than delivery. Share the success criteria at the start, use mini-whiteboards and hinge questions through the middle so every student's work is visible at once, and students finish by marking their own work against a displayed scheme.

Name the skill to students: *"Today you are managing your own accuracy."*

EAL Scaffold

Support Type	Description
Place-Value Grid	Pre-drawn grid (128 to 1) for binary conversions
Step Lists	Repeated-division method as a written step list

Support Type	Description
Sentence Frame	"To convert [binary/hexadecimal] to decimal, I multiply each digit by its place value, which is _____, and sum the results."

2. Lesson Sequence (One-Hour Block)

Part 1: Do Now (Retrieval) — ~10%

Activity: Three questions recalling prior learning:

Question	Source
1. A computer's hardware works with two states. State the two states and give one reason hardware uses two rather than ten	[A1.1.1]
2. How many bits are in one byte, and how many bytes are in one kilobyte when a kilobyte is 2^{10} ?	[A1.1.7] / [A1.1.4]
3. Give one reason a file might be compressed before it is stored or sent	[A1.1.8]

Part 2: Main Sequence — ~60%

Phase 1: Share and Unpack the Criteria (~10 mins)

Display the "I Can" Criteria and model one conversion in each direction from the deck worked examples.

Mark as you go so students see that **the method earns marks as well as the final answer.**

Quick Decimal Place-Value Reminder:

Place	...	1000s	100s	10s	1s
Power of 10	...	10^3	10^2	10^1	10^0

Binary Place-Value Grid:

Place	128	64	32	16	8	4	2
Power of 2	2^7	2^6	2^5	2^4	2^3	2^2	2^1

Hexadecimal Place-Value Grid:

Place	256s	16s	1s
Power of 16	16^2	16^1	16^0

Decimal is the conversion partner rather than a method in its own right.

Phase 2: Generate Visible Evidence (~15 mins)

Activity: Students work conversions on mini-whiteboards while you pose **hinge questions** at the two known decision points.

Hinge Question 1 — Grouping Bits from the Right:

| "When converting binary to hexadecimal, from which end do you group the bits?"

Answer: From the **right**.

Common Error: Grouping from the left. Demonstrate why this fails:

Error (grouping from left)	Correct (grouping from right)
101101 → 101 101	101101 → 10 1101
Adds leading zeros incorrectly	Correct 4-bit grouping: 0010 1101

Hinge Question 2 — Reading Place Value from the Right:

| "Which end is the 1s place in a binary number?"

Answer: The **rightmost** position (bit 0, worth 1).

Common Error: Reading from the wrong end.

Number	Correct Value (rightmost = 1s)	Incorrect (leftmost = 1s)
1010	8 + 2 = 10	1 + 4 = 5

Phase 3: Responsive Self-Check (~5 mins)

Task: Students convert a short set, then check each answer by reversing the conversion against a displayed mark scheme.

| Target the specific error the boards surfaced rather than reteaching everything.

Part 3: Deliberate Practice — ~20%

Activity: Independent or paired work on a conversion set spanning **all six directions**:

Direction	Example
Binary → Decimal	101101 → ?
Decimal → Binary	45 → ?
Hexadecimal → Decimal	2F → ?
Decimal → Hexadecimal	206 → ?
Binary → Hexadecimal	11011110 → ?
Hexadecimal → Binary	3A → ?

Task: Students mark against the criteria and record:

- Which method they used
- How many they got right first time

| This feeds the self-management focus.

Part 4: Exit Check — ~10%

Short Set Tied Directly to A1.2.1:

| "Convert 206 to binary and to hexadecimal showing your working, and convert 2F to decimal."

Answers:

Conversion	Working	Answer
206 → Binary	$128 + 64 + 8 + 4 + 2$	11001110
206 → Hexadecimal	$206 \div 16 = 12$ remainder 14	CE
2F → Decimal	$(2 \times 16) + 15$	47

| Readable in minutes and used to plan the start of the next lesson.

3. Methods for Conversion

Method 1: Place-Value Method (Subtraction)

Example: Convert 45 to binary

Step	Place Value	Can Subtract?	Bit
$45 \geq 32$	Yes ($45 - 32 = 13$)	1	
$13 \geq 16$	No	0	
$13 \geq 8$	Yes ($13 - 8 = 5$)	1	
$5 \geq 4$	Yes ($5 - 4 = 1$)	1	
$1 \geq 2$	No	0	
$1 \geq 1$	Yes ($1 - 1 = 0$)	1	

Result: 101101

Method 2: Repeated Division

Example: Convert 45 to binary

Step	Division	Quotient	Remainder
1	$45 \div 2$	22	1
2	$22 \div 2$	11	0
3	$11 \div 2$	5	1
4	$5 \div 2$	2	1
5	$2 \div 2$	1	0
6	$1 \div 2$	0	1

Read remainders from bottom to top: 101101

Conversion Cheat Sheet

Binary	Hexadecimal	Decimal
0000	0	0
0001	1	1
0010	2	2
0011	3	3
0100	4	4
0101	5	5
0110	6	6
0111	7	7
1000	8	8
1001	9	9
1010	A	10
1011	B	11
1100	C	12
1101	D	13
1110	E	14
1111	F	15

4. Differentiation & Extension

Support (Scaffolds That Can Be Removed)

Support Strategy	Description
Pre-Drawn Place-Value Grid	128 to 1; start with values under 16, then under 256, then remove the grid
Repeated-Division Step List	For students who lose track of the place-value subtraction
Nibble Conversion Table	Binary ↔ hexadecimal reference table

Challenge (Deeper or Wider)

Challenge Strategy	Description
Two-Byte Values	Convert values up to 65535 without drawing the grid
Spot the Error	Students find and correct a wrong conversion

Challenge Strategy	Description
Reasoning Question	"Why does one hexadecimal digit always map to exactly four bits?"

5. Example Hinge Questions

Question	Purpose
"Why is it necessary to convert information into discrete units for computers to process it?"	Checks understanding of digital data
"What are the key differences between decimal, binary, and hexadecimal? Why do we need all three?"	Checks comprehension of purpose and characteristics
"When converting binary to hexadecimal, from which end do you group the bits?"	Targets common grouping error
"Which end is the 1s place in a binary number?"	Targets reading direction error
"What happens if we try to represent a negative number using only unsigned binary?"	Prompts thinking about limitations and signed representations
"Why is hexadecimal useful when working with binary data?"	Assesses understanding of hexadecimal's role as compact representation

6. Learning Objectives (Summary)

Knowledge and Understanding

- Explain the concept of different number systems (binary, hexadecimal, decimal) and their significance in computing
- Identify the base of each number system and the range of digits used
- Explain the relationship between place values and powers of the base

Conversions

- Convert binary integers to decimal and vice versa
- Convert hexadecimal integers to decimal and vice versa
- Convert directly between binary and hexadecimal using four-bit grouping

ATL Skills

Skill	Description
Thinking	Apply critical thinking to analyse and solve conversion problems
Communication	Clearly represent numbers in different notations and explain conversion processes
Self-Management	Monitor own accuracy; choose reliable method for exam conditions

7. Resources

- **Presentation** (Slide Deck)
 - **InThinking eBook** (A1.2.1–A1.2.2 with quiz)
 - **Worksheet:** Conversion set spanning all directions
 - **Worksheet Answers**
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Author's Reflection

"This lesson is the one place in this subtopic where every answer can be easily judged as right or wrong, and that is why I have let assessment lead it rather than exposition. I share the success criteria at the start, put the working on whiteboards where I can read the room, and use the two hinge points—which end place value runs from and which end you group the bits from—to catch the errors that otherwise travel silently into an exam.

Students finish by checking their own answers in reverse, so the accuracy they leave with is theirs to own rather than a mark I have handed them. The one deliberate trade-off is method. I teach both the place-value and the repeated-division routes and then ask students to commit to the one they trust under pressure, rather than prescribing a single approach, because reliability in an exam matters most to the students.

I also held the lesson to integers: the old deck carried fractional place value, and since the statement asks only for whole numbers I have cut it, keeping the load on the conversions that are actually assessed. The slides stay light on worked detail and send students to the eBook for it, so the hour is spent on them converting rather than watching me convert."